

Transport Engineering Group
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The University of Melbourne
Parkville, VIC 3010, Australia

August 23, 2026

Editor-in-Chief
Transportation Research Part C: Emerging Technologies

Dear Editor-in-Chief,

Please consider the manuscript “GEL-Ped: Geometry-encoded learning for pedestrian forecasting under limited data and unseen flow topology” for publication in *Transportation Research Part C: Emerging Technologies*.

The paper addresses a transport-operations question: when should a forecaster retain geometry-encoded structure instead of using a neural model alone? GEL-Ped combines an unrestricted neural expert with a supported residual expert around a ten-coefficient longitudinal–lateral tensor prior. All choices are made on seven complete calibration runs and then frozen for five ordinary corridor runs, three altered geometries, and thirteen perpendicular crossing runs.

The clearest result is conditional rather than universal. On external crossing flow, GEL-Ped lowers velocity RMSE by 1.38% relative to the matched direct network and 1.49% relative to a protocol-matched 2024 goal-stable hybrid; both complete-run comparisons remain significant after Holm correction. Its advantage over the direct network persists from 0.2 to 1.2 s. Once architecture selection is frozen, GEL-Ped is lower for all 127 possible subsets of one to seven calibration runs, including a 14.1% mean reduction with one run. The direct network remains slightly better for familiar-corridor interpolation, and the manuscript states that decision boundary explicitly.

The manuscript fits the journal’s emphasis on emerging methods for transport monitoring, real-time operation, safety, and transferable benchmarking. It engages recent work on interpretable interaction learning, operational navigation, contextual intent prediction, online pedestrian-flow prediction, and hybrid learning–physics crowd models. Public trajectory archives, frozen run splits, a public code repository, automated tests, exact small-sample inference, and an executable manuscript-consistency audit support reproducibility.

This manuscript is original, is not under consideration elsewhere, and all submission statements will be confirmed by the author in Editorial Manager. Thank you for your consideration.

Sincerely,

Amir Ghorbani
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